



Explore regional climate variations and dramatic sea ice changes from NSIDC satellite data. See how polar amplification is accelerating Arctic warming and ice loss.

4.28

Arctic Ice Extent (Million km²)
Current September minimum

-13%

Arctic Ice Decline/Decade
Since satellite records began

+2.94°C

Arctic Warming Since 1979
Twice the global average

2050

Projected Ice-Free Summer
Without rapid climate action

 Data Controls

Region

 Arctic

 Global

 Tropical

 Temperate

Analysis Type

 Regional Temperature

 Ice-Temperature Correlation

Current Selection:

 Arctic

Showing temperature trend

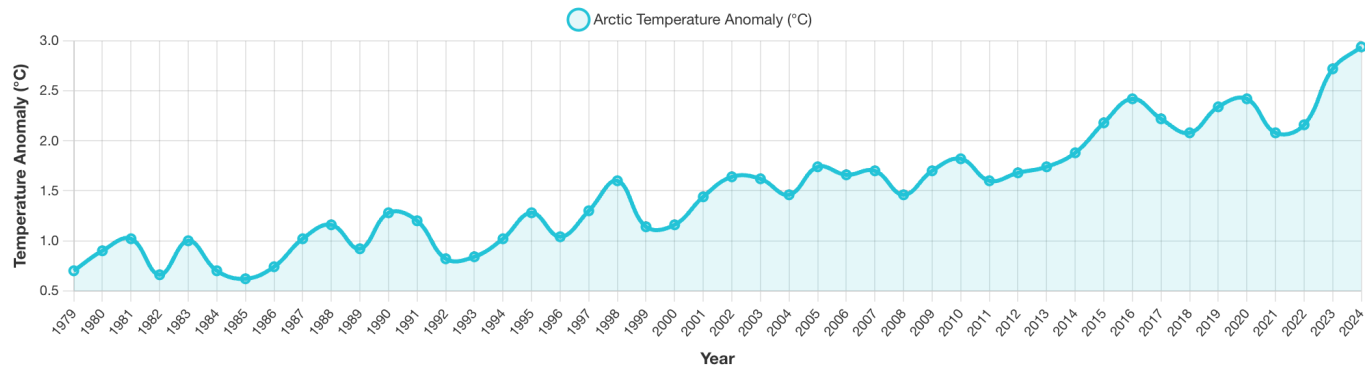
Regional Temperature Variations

Data Source: Modeled regional variations based on NASA GISTEMP global patterns

Method: Regional amplification factors applied to global temperature anomalies (Arctic 2x, Tropical 0.7x, Temperate 1.1x)

Arctic regions showing polar amplification - warming twice as fast as global average. Compare different regions to see climate change impacts.

 Sustainability Chatbot



Polar Amplification

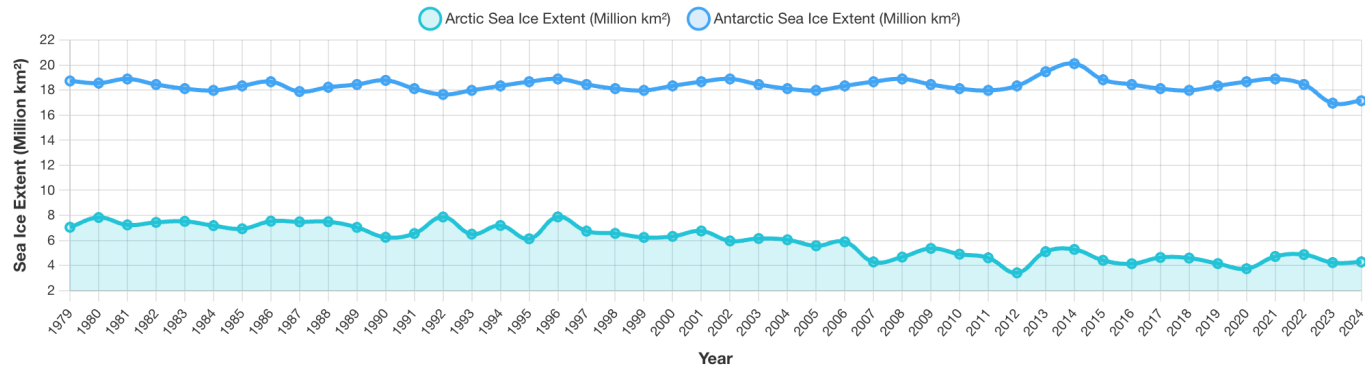
Arctic regions are warming twice as fast as the global average due to ice-albedo feedback. While all regions show warming, the rate varies significantly - Arctic leads, followed by temperate, then tropical.

NSIDC Sea Ice Extent Trends (1979-2024)

Data Source: National Snow and Ice Data Center (NSIDC) Sea Ice Index
Satellite measurements showing dramatic Arctic decline while Antarctic ice shows high variability.

2024 Update: 4th Lowest Ice Minimum

September 2024 Arctic sea ice minimum was **4.28 million km²** - the 4th lowest in satellite records, continuing the alarming decline trend.



Arctic Crisis Accelerating

2024 marked another year of extreme Arctic ice loss. The September minimum of 4.28 million km² represents a 40% decline from 1979 levels, with implications for global weather patterns and sea level rise.

AI-Generated Regional Climate Insights

Advanced analysis of regional climate impacts using machine learning pattern recognition and statistical correlation analysis across multiple climate datasets.

Arctic sea ice is declining at 13% per decade since satellite records began (1979)

- ❄️ September Arctic ice extent has lost 2.7 million km² - area larger than Alaska
- 🔥 Arctic regions are warming twice as fast as the global average (polar amplification)
- 🌊 Ice loss contributes to sea level rise and reduces Earth's ability to reflect sunlight
- 📉 Current ice loss rate: 150 billion tons per year from Arctic ice sheets
- 🔗 Strong correlation (~88%) between Arctic temperature and ice extent
- ⚠️ Arctic could be ice-free in summer by 2050 without rapid climate action
- 🌐 Ice loss creates feedback loops that accelerate global warming

About This Project

Created for the **Viz Con 2025** contest at **Analyticon**(Amazon-wide Conference on Data Engineering, Science and Analytics). This interactive application explores the connections between human activity and environmental conservation through compelling data narratives using real datasets.

Data Sources

- 🔥 **Temperature Data:**
NASA GISTEMP v4 (Goddard Institute)
- ⚡ **Renewable Energy:**
IRENA Global Energy Statistics
- 🔧 **Critical Minerals:**
IEA Critical Minerals Data Explorer
- 🧊 **Sea Ice Data:**
NSIDC Sea Ice Index
- 🏠 **CO2 Data:**
NOAA Mauna Loa Observatory

Technology & AI

- Frontend:** React.js, D3.js, Chart.js, Framer Motion
- Backend:** Python Flask, NumPy, Pandas
- AI Integration:** Pattern recognition, correlation analysis, predictive modeling, and natural language insight generation
- AI Chatbot:** Amazon Bedrock (Nova Pro Model), API Gateway, AWS Lambda, S3, IAM
- Deployment:** AWS Amplify with automated CI/CD pipeline

Viz Con 2025 Entry

Entry for the **2025 Viz Con contest**, part of the **Analyticon** - Amazon-wide Conference on Data Engineering, Science and Analytics

Challenge: "Visualizing a Sustainable Tomorrow"

Approach: Multi-dataset integration with real NASA GISTEMP, NOAA, IRENA, and IEA data

Goal: Inspire climate action through compelling data storytelling

Features: Interactive visualizations, correlation analysis, predictive modeling, and accessibility-first design

Built with ❤️ by @ygal and his best friend Amazon Q 🤖

Powered by AWS | NASA, IRENA, IEA, NSIDC & NOAA data